

DANGHARA, Tajikistan

WMO: 388470

Lat: 38.067N

Lon: 69.283E

Elev: 2195

StdP: 13.57

Time Zone: 5.00 (E05)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.5	21.1	8.9	9.4	25.0	13.6	11.9	34.0	17.2	47.4	14.2	44.9	5.6	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.9	104.3	67.8	101.9	66.6	99.6	65.7	69.9	98.2	68.6	95.5	67.4	93.3	4.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	90.0	74.7	60.2	84.6	75.4	58.9	80.4	75.2	35.3	98.5	34.1	95.2	33.2	93.4	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
14.2	13.3	10.9	10.4	108.2	10.1	2.0	3.1	109.6	-2.8	110.7	-8.5	111.9	-15.8	113.3
			10.8	72.4	9.3	1.4	4.0	73.4	-1.4	74.3	-6.7	75.1	-13.5	76.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	40.4	42.0	55.2	63.5	72.7	81.8	86.9	83.6	75.8	63.8	50.2	42.4	
	DBStd	17.77	7.67	11.01	8.03	6.92	5.25	5.03	3.53	4.19	4.56	7.28	8.27	8.54	
	HDD50	947	304	250	42	5	0	0	0	0	0	3	86	257	
	HDD65	3095	764	644	315	105	5	0	0	1	111	449	701		
	CDD50	5802	5	26	202	410	703	953	1145	1040	774	430	93	21	
	CDD65	2474	0	0	10	60	243	503	680	575	325	73	5	0	
	CDH74	33236	0	6	64	580	2488	6710	10353	8245	3937	819	34	0	
	CDH80	19088	0	0	10	148	1048	3861	6631	5099	2021	266	4	0	
Wind	WSAvg	4.5	4.4	4.9	4.9	4.2	4.4	4.1	4.2	4.8	4.8	4.7	4.1	4.0	
Precipitation	PrecAvg	20.4	2.1	2.8	3.3	3.1	2.4	1.0	0.6	0.3	0.3	0.9	1.7	2.0	
	PrecMax	29.0	3.8	4.8	5.3	7.4	4.7	2.1	1.5	0.8	0.7	2.0	4.4	3.9	
	PrecMin	12.8	0.9	0.9	0.9	1.0	0.4	0.4	0.2	0.1	0.0	0.2	0.5	0.3	
	PrecStd	4.0	0.8	0.9	1.1	1.2	1.1	0.4	0.3	0.2	0.2	0.4	1.0	1.0	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	73.5	80.0	88.8	95.6	104.1	108.1	106.4	98.1	91.8	78.6	66.0
MCWB			45.8	53.3	57.2	62.3	65.6	67.8	70.4	67.3	64.1	60.4	56.1	48.9	
2%		DB	57.3	63.9	74.8	83.9	92.4	100.9	105.4	102.8	96.0	86.4	71.1	61.6	
		MCWB	45.2	49.9	56.0	61.3	65.3	66.2	68.6	66.6	63.0	59.6	53.5	46.8	
5%		DB	54.1	58.6	71.5	80.2	89.3	98.5	103.1	100.6	93.8	82.4	66.5	57.9	
		MCWB	43.9	47.2	54.7	60.3	64.7	65.9	67.6	65.7	62.2	57.7	51.5	45.2	
10%		DB	51.4	55.6	67.7	76.7	86.0	95.9	100.8	98.2	90.8	78.3	62.3	54.9	
		MCWB	42.4	45.7	53.4	59.6	64.4	65.2	66.4	64.8	61.5	56.7	50.2	44.3	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	49.4	54.3	60.5	66.1	69.5	70.2	71.9	69.2	65.7	62.6	57.7	51.2
			MCDB	56.0	68.0	74.1	80.1	88.1	96.1	101.1	100.1	94.3	85.4	73.8	62.0
		2%	WB	47.2	51.2	58.3	64.3	68.0	68.8	70.6	67.6	64.2	60.6	55.4	49.0
			MCDB	54.1	61.6	71.0	78.3	85.2	93.7	100.5	98.8	93.0	81.7	68.1	58.1
	5%	WB	45.4	48.7	56.4	62.7	66.7	67.5	69.4	66.5	63.0	58.9	53.2	47.2	
		MCDB	51.7	57.3	67.3	76.1	83.6	91.8	99.4	97.8	91.4	78.5	63.4	54.5	
	10%	WB	43.8	46.7	54.6	61.1	65.4	66.5	68.1	65.5	61.8	57.4	51.4	45.5	
		MCDB	49.6	54.1	64.5	73.2	81.6	90.6	96.9	96.0	89.3	76.1	59.5	52.3	
	Mean Daily Temperature Range	5% DB	MDBR	14.3	15.3	16.9	19.3	22.4	24.8	26.9	28.0	28.1	24.3	17.5	15.8
			MCDBR	19.4	19.6	21.5	26.0	27.3	28.1	29.7	29.7	31.3	30.1	24.6	20.8
MCWBR			11.1	10.2	10.3	10.7	10.2	9.0	10.9	11.3	13.6	13.5	12.3	10.8	
5% WB		MCDBR	16.2	17.9	19.4	22.5	23.9	25.9	27.3	28.3	30.0	26.2	20.3	18.3	
		MCWBR	10.2	10.1	10.7	10.7	10.5	9.0	10.7	11.3	13.5	12.1	11.0	10.7	
Clear-Sky Solar Irradiance	taub	0.383	0.438	0.498	0.505	0.517	0.549	0.541	0.504	0.464	0.491	0.449	0.394		
	taud	2.093	1.964	1.854	1.878	1.882	1.802	1.809	1.887	1.942	1.845	1.927	2.045		
	Ebn at Noon	240	241	239	248	248	239	239	245	246	220	212	224		
	Edh at Noon	37	49	60	62	63	68	67	61	54	54	44	37		
All-Sky Solar Radiation	RadAvg	672	880	1268	1692	2097	2449	2472	2275	1898	1303	792	601		
	RadStd	76	130	110	173	151	105	77	71	57	112	101	76		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)		N/A	N/A	N/A	N/A	-0.65	-1.22	N/A	N/A	N/A	N/A

Nomenclature: See separate page